Irreversible changes

Nylon jacket

Physical changes are reversible—for example, you can freeze water, and heat can turn the ice into liquid water again. However, many chemical reactions are irreversible because they involve atoms joining together in new ways.

Manmade materials

Chemical reactions can be used to create new materials that don't exist in nature. Nylon, for example, is a fabric made using chemicals from oil. Many types of clothes, from socks to coats, are made of nylon.

Cooking

When food is cooked, heat triggers chemical reactions that change it permanently. When a freshly baked cake cools down, it doesn't turn back into gooey cake mixture.

Baking powder

Baking powder makes cakes light and fluffy. It contains chemicals that react when they're wet to produce bubbles of gas.

A fresh pepper looks plump and brightly colored.

An old pepper darkens and shrivels up as it rots.

Rotting

Rotting food is full of tiny organisms such as a bacteria and fungi. These organisms trigger chemical reactions that break down food molecules, changing them permanently.